

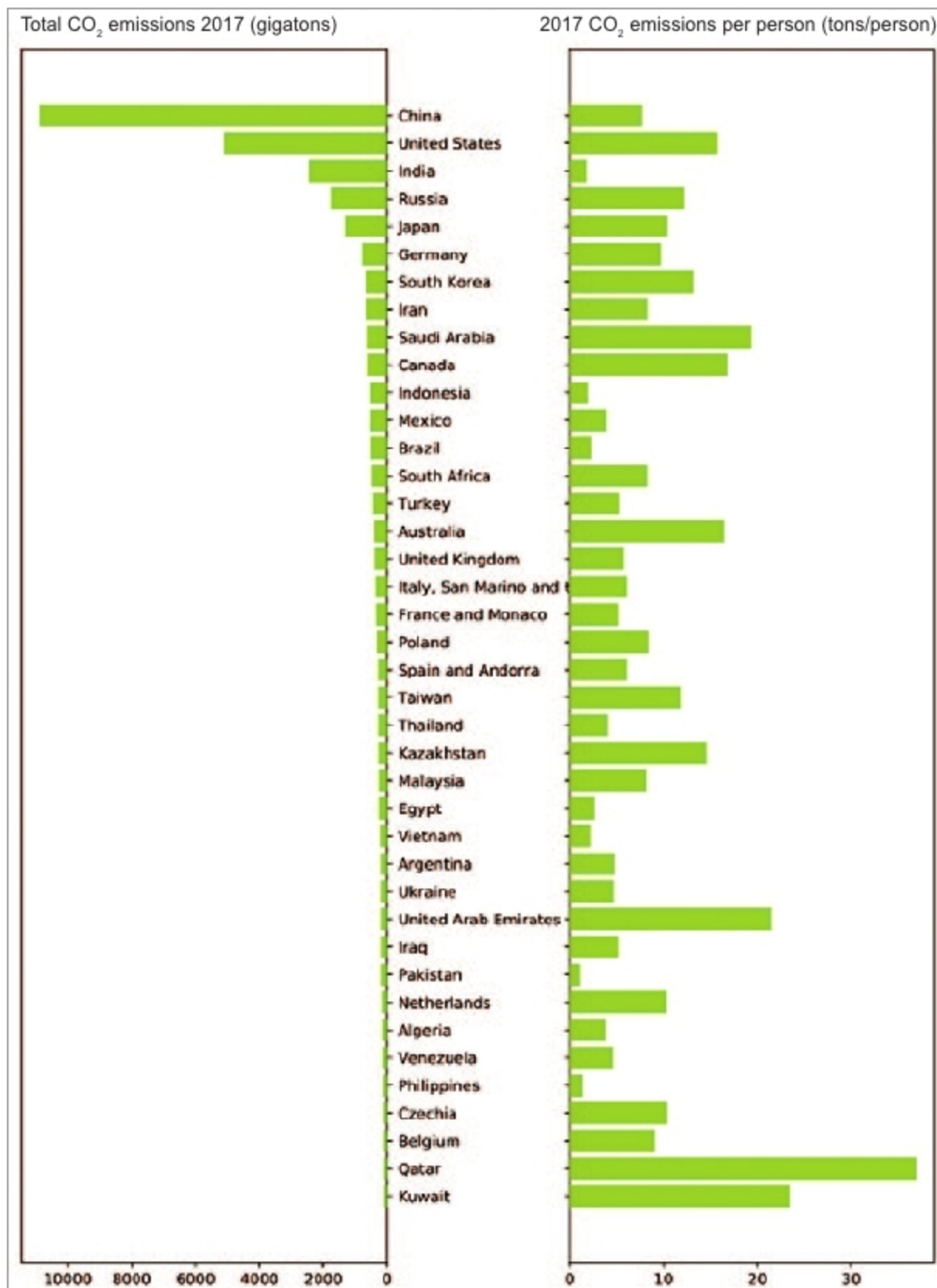


Figure 2: Global carbon dioxide emissions (Credit: Wikipedia.org)

more energy than others. Impact of ICT industries on emission of carbon-dioxide is immense. As shown in Figure 2, India is currently the third largest producer of carbon dioxide.

Urgent solutions required at the level of hardware design management include minimal configuration, adoptive configuration, consolidation by virtualisation, algorithmic efficiency, optimal resource utilisation, optimal data centres, optimal link utilisation, limiting power by reducing cable

length, minimising protocol overhead, protocol for compressed header, green networking, management

of e-waste, air management and cooling management, among others. For ICT scientists and engineers, the challenge will be to design technology and algorithms to minimise particle vibration, travel path and heat loss due to input-output mismatch. Design, operational and transmission related thermal loss are core issues of ICT. This makes production of green ICT a great challenge, although, as parts of its implementation, energy-smart devices, sleep-mode devices, cluster computing, cloud computing, etc are already in place.

Foundation of green ICT was laid as far back as 1992 with the launch of the Energy Star program in the USA. The success of Energy Star motivated other countries to take up the subject for investigation and implementation. Leading countries working on green ICT now include Japan, Australia, Canada and the European Union. Formalisation of green ICT is in fact due to standards proposed by IEEE who has formalised Green Ethernet and 802.3az-enabled devices for green ICT.

Green ICT is a clean-environment-based technology. However, fruitful realisation of green ICT is equally dependent upon awareness in society. Society needs to practice common ethics of 'don't keep computer on, when not needed,' 'don't use Internet as a free tool, but as a valuable tool of necessity only,' 'don't unnecessarily replace devices after devices just because you can afford to' and so on. Without societal responsibility, technology alone cannot ensure achieving the objectives of green ICT. **END** 🐧

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